17/02/2020

1.1 SYSTEMS OF LINEAR EQUATIONS

Let $a_1, a_2, \dots, a_n, b$ be constant numbers

$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$ is called a linear equation in n-unknowns

where $x_1, \dots, x_n$ unknowns are here.

A system of m linear equations in n-unknowns:

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

- $3x_1 + 2x_2 - x_3 = 5$ A system of 2 eq. in 3 unknowns

  $\pi^2 x_1 - \frac{e}{4}x_2^{10x_3} = 6$

$x_1 = s_1, x_2 = s_2, x_n = s_n$ is called a solution if it satisfies all the eq. in

We call a linear system consistent if it has at least one solution, otherwise inconsistent

A linear system is called homogeneous if $b_1 = b_2 = \dots = b_n = 0$, otherwise non-homogeneous

- $3x_1 + 4x_2 = 12$ — 1 eq. non-hom. 2 unk.

  $x_1 = 0 \quad x_2 = 3$

  $x_1 = 4 \quad x_2 = 0$ — consistent

  $x_1 = \pi \quad x_2 = \frac{12 - 3\pi}{4}$

- $3x_1 + 4x_2 = 12$
  $3x_1 + 4x_2 = 0$ Inconsistent because $12 \neq 0$

Non-hom linear systems can be consistent or not.
Hom. linear systems are always consistent. Because $x_1 = x_2 = \dots x_n = 0$ is a solution. This solution is called trivial solution.

Two linear systems are equivalent if they have the same set of solutions.

- $x_1 + x_2 - x_3 = 4$
  and
  $x_1 + x_2 - x_3 = 4$ are equivalent
  $2x_1 + 2x_2 - 2x_3 = 8$

To find a solution, we use the method of elimination

- $x - 3y = -7$ eq 1
  $2x - 6y = 7$ eq 2
  $-2(\text{eq}1) + \text{eq}2 \Rightarrow \begin{cases} x - 3y = -7 \\ 0 = 21 \end{cases}$
  ↳ inconsistent

- $2x - 3y + 2z = 14$ eq 2
  $x + 2y + 3z = 6$ eq 1
  $3x + y - z = -2$ eq 3
  $-2(\text{eq}1) + (\text{eq}2) \to (\text{eq}2)$
  $-3(\text{eq}1) + (\text{eq}3) \to (\text{eq}3)$

  $x + 2y + 3z = 6$ eq. 1
  $-7y - 4z = 2$ eq. 2
  $-5y - 10z = -20$ eq. 3
  $-\frac{1}{5}(\text{eq}3) \to (\text{eq}3)$
  $(\text{eq}3) \leftrightarrow (\text{eq}2)$

  $x + 2y + 3z = 6$ eq 1
  $-5y - 10z = -20$ eq 3
  $y + 2z = 4$ eq 2
  $-7(\text{eq}2) + (\text{eq}3) \to (\text{eq}3)$

  $x + 2y + 3z = 6$ eq 1
  $y + 2z = 4$ eq 2
  $10z = 30$

  $x = 1$
  $z = 3$
  $y = -2$
  } unique solution
  no other sh

• $x+2y-3z=-4$ $\quad -2(eq1)+eq2 \to (eq2)$

$2x+y-3z=4$

$z \to$ any number

$x+2y-3z=-4$ $\quad y=z-4$

$-3y+3z=12 \iff z-y=4$ $\quad x=z+4$

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1.2 MATRICES

$$A=\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & & & & \\ a_{m1} & \cdots & \cdots & \cdots & a_{mn} \end{bmatrix}$$

$m \times n$ matrix

$A=[a_{ij}]$

If $m=n$ square matrix

If $n=1$ m-vector

If they have the same size and same (i,j)-entry for each i,j two matrices are equal.

- Matrix Addition

$A=[a_{ij}]_{m \times n}$ $\quad B=[b_{ij}]_{m \times n}$ $\quad C=A+B$ $\quad C=[c_{ij}]$, $c_{ij}=a_{ij}+b_{ij}$

If does not have the same size A+B undefined

- Scalar Product

$c \in R$, $A_{m \times n}$ $\quad cA=[ca_{ij}]$ • $A=\begin{bmatrix} 1 & 4 \\ 5 & 7 \end{bmatrix}$ $\quad -3A=\begin{bmatrix} -3 & -12 \\ -15 & -21 \end{bmatrix}$

- Transpose

$A_{m \times n}=[a_{ij}]$ $\quad A^T=[a_{ji}] \to n \times m$ matrix • $A=\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$

$A^T=\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

• $A=\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ $\quad B^T=\begin{bmatrix} 0 & 7 & 1 \\ 3 & -1 & 2 \end{bmatrix}$ $\quad A+5B^T=\begin{bmatrix} 1 & 37 & 8 \\ 19 & 0 & 16 \end{bmatrix}$

$B=\begin{bmatrix} 0 & 3 \\ 7 & -1 \\ 1 & 2 \end{bmatrix}$ $\quad 5B^T=\begin{bmatrix} 0 & 35 & 5 \\ 15 & -5 & 10 \end{bmatrix}$

$A+5B^T=?$

- Linear Combinations

If $A_1, A_2, \dots A_k$ are $m \times n$ matrices

$c_1, c_2, \dots c_k$ are scalars

$c_1A_1 + c_2A_2 + \dots + c_kA_k \rightarrow$ linear combination

1.3 MATRIX MULTIPLICATION

- Dot product of two n-vectors

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = a_1b_1 + a_2b_2 + \dots + a_nb_n$$

- Matrix product

$A_{m \times p}$ matrix, $B_{p \times n}$ matrix $\quad A = [a_{ij}]$, $B = [b_{ij}]$

$$AB = C = [c_{ij}] \qquad c_{ij} = \sum_{k=1}^{p} a_{ik} b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ip}b_{pj}$$

$\downarrow$ $m \times n$ matrix

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ a_{21} & & & \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mp} \end{bmatrix}_{m \times p} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & & & \\ \vdots & & & \\ b_{p1} & b_{p2} & \cdots & b_{pn} \end{bmatrix}_{p \times n} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ & & & \\ & & & \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{bmatrix}_{m \times n}$$

- $c_{12} = (\text{Row } 1 \text{ of } A)^T \cdot (\text{Col } 2 \text{ of } B)$
- $A_{3 \times 2}$, $B_{3 \times 3}$ $\quad AB$ is undefined $\quad BA$ is defined $\rightarrow$ 3×2 matrix

- $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 0 \\ 7 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 4 & 5 \\ -2 & 0 \end{bmatrix} = \begin{bmatrix} [1\ 2\ 3]^T \cdot \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix} & [1\ 2\ 3]^T \cdot \begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix} \\ [0\ 5\ 0]^T \cdot \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix} & [0\ 5\ 0]^T \cdot \begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix} \\ [7\ -1\ 0]^T \cdot \begin{bmatrix} 0 \\ 4 \\ -2 \end{bmatrix} & [7\ -1\ 0]^T \cdot \begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix} \end{bmatrix}$

- $\begin{bmatrix} 2 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 3 & 4 \\ 1 & 0 & -1 \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} & \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \end{bmatrix} & \begin{bmatrix} 2 \\ 1 \end{bmatrix} \begin{bmatrix} 4 \\ -1 \end{bmatrix} \\ \begin{bmatrix} 0 \\ -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} & \begin{bmatrix} 0 \\ -1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ -1 \end{bmatrix} \begin{bmatrix} 4 \\ -1 \end{bmatrix} \end{bmatrix}$

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If $A_{5\times4}$ $B_{4\times5} \Rightarrow$ AB is defined but BA is not defined

$$\left.\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \end{aligned}\right\} x_1, x_2, x_3 \text{ are unknowns} \rightarrow A\underset{\sim}{x} = \underset{\sim}{b}$$

write as a matrix equation

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \qquad \underset{\sim}{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad \underset{\sim}{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

coefficient matrix

$$A\underset{\sim}{x} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \underset{\sim}{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \end{bmatrix}$$

- $3x_1 + 2x_2 = 5$
  $4x_1 = 7$
  $3x_1 - x_2 = 0$

$\rightarrow$ A $\begin{bmatrix} 3 & 2 \\ 4 & 0 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \underset{b}{\begin{bmatrix} 5 \\ 7 \\ 0 \end{bmatrix}}$

1.4 ALGEBRAIC PROPERTIES OF MATRIX OPERATIONS

- Matrix additions

$A, B, C \rightarrow n \times n$ matrices

i) $A+B = B+A$ ii) $A+(B+C) = A+B+C$

iii) There is a unique matrix $0_{m \times n}$ such that $A+0 = 0+A = A$ $0 \rightarrow$ zero matrix

iv) $A + (-1)A = 0$

- Multiplications

i) $(AB)C = A(BC)$

$A_{m \times n}$ $B_{n \times p}$ $C_{p \times r}$

$(AB)C$: $(m \times p)(m \times r)$ $A(BC)$: $(n \times r)(m \times r)$

ii) $A(B+C) = AB+AC$ $(A+B)C = AC+BC$

- Properties of Scalar Multiplications

If r, s are number A, B are matrices of suitable sizes.

(i) $r(sA) = (rs)A$ (iii) $r(A+B) = rA+rB$

(ii) $(r+s)A = rA+sA$ (iv) $A(rB) = r(AB)$

- Transpose

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

If r is a scalar

A, B matrices of suitable sizes.

(i) $(A^T)^T = A$ (ii) $(A+B)^T = A^T + B^T$ (iii) $(AB)^T = B^T A^T$

$A_{m \times n}$ $B_{n \times p} \rightarrow (AB)_{m \times p}$

$A^T_{n \times m}$ $B^T_{p \times m} \rightarrow$

$(AB)^T \neq A^T B^T$

- $A \neq 0, B \neq 0 \quad AB = 0$ is possible
- $AB = AC$ but $B \neq C$ may happen

1.5 SPECIAL TYPES OF MATRICES

1 $A_{n \times n}$ → square matrix

2. $A_{n \times n}$ is called a diagonal matrix if $a_{ij} = 0$ for $i \neq j$

ex. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

3. A diagonal matrix is called a scalar matrix if all its diagonal entries are equal

ex. $\begin{bmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$ → scalar matrix

4. A scalar matrix is called identity matrix if all its diagonal entries are 1. (I_n)

$I_n \cdot A = A \qquad I = [\delta_{ij}] \qquad \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$

Powers of matrices

$A^2 = \underset{m \times n}{A} \underset{m \times n}{A}$

$A_{n \times n} \Rightarrow A^0 = I_n \quad A^1 = A \quad A^2 = A \cdot A \quad A^m = \underbrace{A \cdots A}_{m \text{ times}}$

ex. $A = \begin{bmatrix} 1 & 5 \\ 0 & 2 \end{bmatrix} \quad A^2 = \begin{bmatrix} 1 \cdot 1 + 5 \cdot 0 & 1 \cdot 5 + 5 \cdot 2 \\ 0 \cdot 1 + 2 \cdot 0 & 0 \cdot 5 + 2 \cdot 2 \end{bmatrix} = \begin{bmatrix} 1 & 15 \\ 0 & 4 \end{bmatrix}$

for non-negative integers p, q

$A^p A^q = A^{p+q} \qquad (A^p)^q = A^{pq} \qquad (AB)^p \neq (BA)^p$

- $n \times n$ matrix is called upper triangular if $a_{ij} = 0$ for $i > j$

$$\begin{bmatrix} 1 & 1 & 2 & 0 \\ 0 & 3 & 0 & 4 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

↳ must be zero

- $n \times n$ matrix is called lower triangular if $a_{ij} = 0$ for $j > i$

- A square matrix is symmetric matrix if $A = A^T$

$$\begin{bmatrix} 1 & 3 \\ 3 & 0 \end{bmatrix}^T = \begin{bmatrix} 1 & 3 \\ 3 & 0 \end{bmatrix}$$

skew symmetric if $A = -A^T$ $\quad \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}^T = \begin{bmatrix} 0 & 3 \\ -3 & 0 \end{bmatrix}$

3×3 symmetric $\begin{bmatrix} a & d & e \\ d & b & f \\ e & f & c \end{bmatrix}$ $\quad$ 3×3 skew-symmetric $\begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$

2×2 symmetric $\begin{bmatrix} a & c \\ c & b \end{bmatrix}$

Any square matrix can be ~~written~~ written as a unique sum of symmetric and skew-symmetric matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} d & x \\ x & d \end{bmatrix} + \begin{bmatrix} 0 & y \\ -y & 0 \end{bmatrix}$$

$$x + y = b$$
$$x - y = c$$

$$x = \frac{b+c}{2}$$
$$y = \frac{b-c}{2}$$

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$A_{n\times n}$ square matrix if there exists an $n\times n$ matrix B such that

$$AB = BA = I_n$$

then we call B the inverse of A and denote $B = A^{-1}$

If A has an inverse then we call A a non-singular or invertible matrix otherwise it is called a singular matrix

1. The matrix inverse, if it exists is unique

proof.

Suppose A has two inverses B and C

$$AB = BA = I \qquad AC = CA = I$$

$$B = BI = B(AC)$$
$$= (BA)C = IC = C$$

- Find the inverse of $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ if it exists.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} x+2z & y+2w \\ 3x+4z & 3y+4w \end{bmatrix}$$

$$\begin{array}{ll} x+2z = 1 & y+2w = 0 \\ 3x+4z = 0 & 3y+4w = 1 \\ \hline x = -2 & y = 1 \\ z = 3/2 & w = \frac{-1}{2} \end{array}$$

$$A^{-1} = \begin{bmatrix} -2 & 1 \\ 3/2 & \frac{-1}{2} \end{bmatrix}$$

2. If A and B are invertible then so is AB

proof.

$$(AB)x = I$$

$$ABB^{-1} = AI = A$$

$$(AB)(B^{-1}A^{-1}) = AA^{-1} = I$$

$$(B^{-1}A^{-1})(AB) = B^{-1}IB = B^{-1}B = I$$

in general $(AB)^{-1} \neq A^{-1}B^{-1}$

$\begin{bmatrix} 0 & 1 & 5 \\ 1 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ the leading 1 to the left of the leading 1 above not in REF (1)

$\begin{bmatrix} 1 & 2 & 5 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ in REF

$\begin{bmatrix} 1 & 0 & 0 & 5 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ in RREF

3. If $A_1, \dots, A_n$ invertible then

$$(A_1 \dots A_n)^{-1} = A_n^{-1} A_{n-1}^{-1} \dots A_2^{-1} A_1^{-1}$$

4. If A is invertible so is A^{-1} and $(A^{-1})^{-1} = A$

proof

$A^{-1}X = I$ A is inverse of A^{-1} $(A^{-1})^{-1} = A$

5. If A is invertible so is A^T and $(A^T)^{-1} = (A^{-1})^T$

proof

$$A^T (A^{-1})^T = (A^{-1}A)^T = I^T = I$$

$$(A^{-1})^T A^T = I$$

Suppose A is invertible and we are solving

$$A\underset{\sim}{x} = \underset{\sim}{b} \quad \text{for } \underset{\sim}{x}$$

$$A^{-1}A\underset{\sim}{x} = \underset{\sim}{b}A^{-1} \implies \underbrace{\underset{\sim}{x} = A^{-1}\underset{\sim}{b}}_{\text{solution}}$$

ex

$$x + 2y = 8$$
$$3x + 4y = 6$$

$$A\underset{\sim}{x} = \underset{\sim}{b}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$$

$$\underbrace{A^{-1}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}}_{I}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1}\begin{bmatrix} 8 \\ 6 \end{bmatrix} = \begin{bmatrix} -10 \\ 9 \end{bmatrix}$$

$\hookrightarrow A^{-1} = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}$

check $x = -10$, $y = 9$ satisfies:

$$x + 2y = 8$$
$$3x + 4y = 6$$

- Suppose $(A^2)^{-1} \cdot \underset{\sim}{x} = \underset{\sim}{b}$ with $A^{-1} = \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}$, $\underset{\sim}{b} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

$$A^{-1} A^2 \underset{\sim}{x} = A^{-1} \underset{\sim}{b}$$

$$I A \underset{\sim}{x} = A^{-1} \underset{\sim}{b}$$

$$A \underset{\sim}{x} = A^{-1} \underset{\sim}{b} \Rightarrow A^{-1} A \underset{\sim}{x} = A^{-1}(A^{-1} \underset{\sim}{b}) \Rightarrow \underset{\sim}{x} = A^{-1}(A^{-1} b)$$

$$= A^{-1} \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} -3 \\ -6 \end{bmatrix}$$

02/02/2020

CHAPTER-2 SOLVING LINEAR SYSTEMS

2.1 Echelon Form

A_{mxn} is in row echelon form (REF) if

1) All zero rows, if any, are at the bottom of the matrix
2) The first nonzero entry from left of a nonzero row is 1. This entry called a leading one of its row.
3) For each nonzero row, the leading one appears to the right and below of any leading ones in ~~preceding~~ above rows.

If is additive A satisfies

4) If a column contains a leading one then all other entries in that column are zero

Then we say A is in ~~reduced REF~~ RREF

$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 2 & 3 \end{bmatrix}$ doesn't satisfy (1)

$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 5 \\ 0 & 0 & 0 \end{bmatrix}$ not REF (2)

$\begin{bmatrix} 12 \\ 13 \end{bmatrix}$ not in REF (3)

$\begin{bmatrix} 0 & 1 & 5 \\ 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ this leading 1 is to the left of the leading 1's above not in REF (3)

$\begin{bmatrix} 1 & 2 & 5 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is REF

$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ is REF (4) not in RREF

$\begin{bmatrix} 1 & 0 & 0 & 5 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ in RREF

$$\begin{bmatrix} 1 & 0 & * \\ 0 & 1 & * \end{bmatrix}$$

Entries below leading one are zero is REF

Entries below and above leading 1 are zero is RREF

- Elementary Row Operations

1. $R_i \leftrightarrow R_j$ swap row i with row j

2. $kR_i \to R_i$ $(k \neq 0)$ replace row i by k · (row i)

3. $kR_j + R_i \to R_i$ replace row i by k · (row j) + (row i)

$A_{m\times n}$, $B_{m\times n}$ If B can be obtained from A by a sequence of elementary row operations then we say B is row equivalent to A.

$$A = \begin{bmatrix} 1 & 5 & 3 \\ 2 & 0 & 9 \end{bmatrix} \qquad B = A_{2R_2 \leftrightarrow R_1} = \begin{bmatrix} 2 & 0 & 9 \\ 1 & 5 & 3 \end{bmatrix} \qquad C = B_{2R_1 \to R_1} = \begin{bmatrix} 4 & 0 & 18 \\ 1 & 5 & 3 \end{bmatrix}$$

$$D = C_{3R_2 + R_1 \to R_1} = \begin{bmatrix} 4+3 & 0+15 & 18+3 \\ 1 & 5 & 3 \end{bmatrix}$$

A, B, C, D row equivalent to each other.

Theorem If a matrix A is row equivalent to B then B is row eq. A.

Theorem Every matrix is row equivalent to a matrix is REF.

Moreover, every matrix is row equivalent to a unique matrix is RREF

- $A = \begin{bmatrix} 2 & 4 & 10 \\ 3 & 5 & 0 \end{bmatrix}$ $\quad B = A_{\frac{1}{2}R_1 \to R_1} \begin{bmatrix} 1 & 2 & 5 \\ 3 & 5 & 0 \end{bmatrix}$

$$C = B_{-3R_1 + R_2 \to R_2} = \begin{bmatrix} 1 & 2 & 5 \\ 3-3 & 5-6 & 0-15 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 5 \\ 0 & -1 & -15 \end{bmatrix}$$

$$D = C_{-R_2 \to R_2} = \begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 15 \end{bmatrix}$$

Find a matrix which is RREF and equivalent to A.

$$E = \underset{-2R_2+R_1\to R_1}{D} = \begin{bmatrix} 1 & 0 & -25 \\ 0 & 1 & 15 \end{bmatrix}$$ is RREF

$$F = \underset{-R_2+R_1\to R_1}{D} = \begin{bmatrix} 1 & 1 & -10 \\ 0 & 1 & 15 \end{bmatrix}$$ is REF

D and F are both in REF and row eq. to A.

REF of a matrix isn't unique

RREF is unique

proof of theorem:

Let $A_{m\times n}$ matrix.

1) Let $i = 1$

2) Let the first non-zero entry in COL, at or below Row_i be in Row_j. If there are none go to the last step.

3) $Row_i \leftrightarrow Row_j$

4) By choosing suitable k and using the operation $kR_i \to R_i$ make the leading entry of ROW_i to be 1.

5) By choosing suitable k and using the operations $kR_i + R_j \to R_j$ make the entries below the leading 1 to be zero.

6) If $i = m$ then stop. If not increase i by 1 and go to 2nd step.

- $x + 2y + 3z = 6$
  $2x - 3y + 2z = 14$
  $3x + y - z = -2$

Solve for x, y, z.

write in augmented form

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & -3 & 2 & 14 \\ 3 & 1 & -1 & -2 \end{array}\right]$$

Bring to REF using elementary row operations

$-2R_1 + R_2 \to R_2$

$-3R_1 + R_3 \to R_3$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -7 & -4 & 2 \\ 0 & -5 & -10 & -20 \end{array}\right]$$

$R_2 \leftrightarrow R_3$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -5 & -10 & -20 \\ 0 & -7 & -4 & 2 \end{array}\right]$$

$-\frac{1}{5}R_2 \to R_2$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & -7 & -4 & 2 \end{array}\right]$$

$7R_2 + R_3 \to R_3$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 10 & 30 \end{array}\right]$$

$\frac{1}{10} R_3 \to R_3$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right] \text{ REF}$$

$x + 2y + 3z = 6$

$y + 2z = 4$

$z = 3 \Rightarrow y = -2, \; x = 1$

03/02/2020

- Solving Linear Systems

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + \dots + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n = b_m \end{cases}$$

System of m linear equations in n unknowns.

$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ a_{m1} & \dots & a_{mn} \end{bmatrix}_{m \times n} \qquad \vec{x} = \begin{bmatrix} x_1 \\ i \\ x_n \end{bmatrix}_{n \times 1}$$

$$\vec{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}_{m \times 1}$$

$$A\vec{x} = \vec{b}$$

To solve form the augmented matrix

$$[A|b]_{m \times (n+1)} \xrightarrow{\text{using elementary row op.}} \begin{cases} \nearrow \text{REF} \to \text{solve by backward substitution (gaussian elimination)} \\ \searrow \text{RREF} \to \text{solve by backward substitution (gauss-jordan reduction)} \end{cases}$$

- $x + 2y + 3z = 6$
  $2x - 3y + 2z = 14$
  $3x + y - z = -2$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & -3 & 2 & 14 \\ 3 & 1 & -1 & -2 \end{array}\right] \xrightarrow{\text{REF}} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$z = 3$

$y + 2z = 4$

$x + 2y + 3z = 6$

gauss-jordan reduction

$-2R_3 + R_2 \to R_2$

$-3R_3 + R_1 \to R_1$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & -3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$-2R_2 + R_1 \to R_1$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{array}\right] \Rightarrow \begin{array}{l} x = 1 \\ y = -2 \\ z = 3 \end{array}$$

- $\left[\begin{array}{ccccc|c} x_1 & x_2 & x_3 & x_4 & x_5 & \\ 1 & 2 & 3 & 4 & 5 & 6 \\ 0 & 1 & 2 & 3 & -1 & 7 \\ 0 & 0 & 1 & 2 & 3 & 1 \\ 0 & 0 & 0 & 1 & 2 & 9 \end{array}\right]$

$x_5 = r$ (any real number)

$x_4 = 9 - 2r$

$x_3 = 7 - 2(9-2r) = r - 11$

$x_2 = 7 - 2(r-11) - 3(9-2r) + r = 5r + 2$

$x_1 = -1 - 10r$

Solution: $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = r \begin{bmatrix} -10 \\ 5 \\ 1 \\ 2 \\ 1 \end{bmatrix}$ infinite sh.

- $\left[\begin{array}{ccccc|c} x_1 & x_2 & x_3 & x_4 & x_5 & \\ 1 & 1 & 2 & 0 & -5/2 & 2/3 \\ 0 & 0 & 0 & 1 & 1/2 & 1/2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$

$x_2 = s$

$x_3 = t$

$x_5 = r$

$x_1 + x_2 + 2x_3 - \frac{5}{2}x_5 = \frac{2}{3}$

$x_1 = \frac{2}{3} - s - 2t + \frac{5}{2}r$

$x_4 + \frac{1}{2}x_5 = \frac{1}{2} \Rightarrow x_4 = \frac{1}{2} - \frac{1}{2}r$

$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2/3 \\ 0 \\ 0 \\ 1/2 \\ 0 \end{bmatrix} + s\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + r\begin{bmatrix} 5/2 \\ 0 \\ 0 \\ -1/2 \\ 1 \end{bmatrix}$ s, t, r any real number

- Find the quadratic polynomial passing from the points $(1, -5), (-1, 1), (2, 7)$

$p(x) = ax^2 + bx + c$

$a + b + c = -5$
$a - b + c = 1$
$4a + 2b + c = 7$

$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & -5 \\ 1 & -1 & 1 & 1 \\ 4 & 2 & 1 & 7 \end{array}\right]$

$-R_1 + R_2 \to R_2$
$-4R_1 + R_3 \to R_3$

$\left[\begin{array}{ccc|c} 1 & 1 & 1 & -5 \\ 0 & -2 & 0 & +6 \\ 0 & -2 & -3 & -27 \end{array}\right]$

$-\frac{1}{2}R_2 \to R_2$

$\left[\begin{array}{ccc|c} 1 & 1 & 1 & -5 \\ 0 & 1 & 0 & -3 \\ 0 & -2 & -3 & 27 \end{array}\right]$

$2R_2 + R_3 \to R_3$

$\left[\begin{array}{ccc|c} 1 & 1 & 1 & -5 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & -3 & 21 \end{array}\right]$

$-3c = 21 \Rightarrow c = -7$

$b = -3$

$a + b + c = -5 \Rightarrow a = 5$

$p(x) = 5x^2 - 3x - 7$

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Special case

$\vec{b} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \Rightarrow$ homogenous equation $\Rightarrow$ always at least one solution $\rightarrow$ the trivial solution $\rightarrow \vec{x} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{n \times 1}$

- $3x_1 - 2x_2 + x_3 = 0$
  $2x_1 - 3x_2 - 7x_3 = 0$

  $x_1 = x_2 = x_3 = 0 \Rightarrow$ trivial solution

Theorem: For a homogenous system

if $m < n$ (m: number of rows / equations; n: number of unknowns / columns)

then there are infinitely many solutions

- $x_1 + 2x_2 - 3x_3 = 0$
  $-2x_1 + x_2 + x_3 = 0$

$\begin{bmatrix} 1 & 2 & -3 & 0 \\ -2 & 1 & 1 & 0 \end{bmatrix}$ $2R_1 + R_2 \rightarrow R_2$ $\begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 5 & -5 & 0 \end{bmatrix}$ $\frac{1}{5} R_2 \rightarrow R_2$ $\begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix}$

$x_3 = t$ any R number

$x_2 - x_3 = 0 \quad x_2 = t$

$x_1 = t$

$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ infinitely many solutions

Let k = # of leading 1's then $k \leq m$ (each row have at most one leading 1)

There are $n - k > 0$ free variables

- $x + y + z + w = 0$
  $x + w = 0$
  $x + 2y + z = 0$

$\begin{bmatrix} 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 \end{bmatrix}$ $-R_1 + R_2 \rightarrow R_2$, $R_1 - R_3 \rightarrow R_1$ $\begin{bmatrix} 0 & -1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 0 & -1 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 \end{bmatrix}$

- Balance the chemical equation

$$x\,NaOH + y\,H_2SO_4 \rightarrow z\,Na_2SO_4 + w\,H_2O$$

Na: $x - 2z = 0$

O: $x + 4y - 4z - w = 0$

H: $x + 2y - 2w = 0$

S: $y - z = 0 \Rightarrow y = z$

$y = t \rightarrow$ free parameter

$x = 2t$

$w = x = 2t$

$z = y = t$

$x - 2y = 0$

$x - w = 0 \Rightarrow x = w$

$x + 2y - 2w = 0$

$x - 2y = 0$

$x + 2y - 2x = 0 \Rightarrow 2y - x = x + y$

$$\left[\begin{array}{cc|c} 1 & -2 & 0 \\ -1 & 2 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

$$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$$

- Determine of values of a such that

$x + y - z = 2$

$x + 2y + z = 3$

$x + y + (a^2 - 5)z = a$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 1 & 2 & 1 & 3 \\ 1 & 1 & a^2-5 & a \end{array}\right]$$

$-R_1 + R_2 \rightarrow R_2$

$-R_1 + R_3 \rightarrow R_3$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & a^2-4 & a-2 \end{array}\right]$$

a) If $a^2 - 4 = 0$ but $a = -2$ then there is no solution

$0x + 0y + 0z = -4$

b) $a^2 - 4 \neq 0$ then there is a unique solution a

c) If $a = 2$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

no leading 1 for z - variable

$z = t \Rightarrow$ free par.

$y = 2t + 1$

$x = 1 - t$

∞ solutions

- Find an equation relating a, b and c so that the linear system

$2x + 2y + 3z = a$

$3x - y + 5x = b$

$x - 3y + 2z = c$

Relations between homogeneous system - and non-homogeneous system

Suppose

$A\vec{x}_h = \vec{0}$ $\quad A\vec{x}_p = \vec{b}$ $\quad$ then $\quad A(\underbrace{\vec{x}_h + \vec{x}_p}_{\text{solutions of non-hom. system}}) = A\vec{x}_h + A\vec{x}_p = \vec{0} + \vec{b} = \vec{b}$

2.3 Elementary Matrices Finding A^{-1}